* Rank 0 tensor.

$$[T]$$

$$T = \mathbb{T}_0^{(1)}$$

* Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} T_{xx} & 0 & 0 \\ 0 & T_{xx} & 0 \\ 0 & 0 & T_{zz} \end{bmatrix}$$

$$T_{xx} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(1)}$$
$$T_{zz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_u^{(1)}$$

* Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} T_{xxx} & 0 & 0 \\ -T_{xxx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & -T_{xxx} & 0 \end{bmatrix}$$

$$T_{xxx} = \mathbb{T}_{3c}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ T_{yzx} & 0 & 0 \\ 0 & -T_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzx} = 3\mathbb{M}_u^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} T_{yzx} & 0 & 0 \\ 0 & T_{yzx} & 0 \\ 0 & 0 & T_{xyz} \end{bmatrix}$$

$$T_{yzx} = \mathbb{M}_0^{(1)} - \mathbb{M}_u^{(2)}$$
$$T_{xyz} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_u^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} T_{xxxx} & T_{xxyy} & T_{xxzz} & T_{xxyz} & 0 & 0 \\ T_{xxyy} & T_{xxxx} & T_{xxzz} & -T_{xxyz} & 0 & 0 \\ T_{xxzz} & T_{xxzz} & T_{zzzz} & 0 & 0 & 0 \\ T_{xxyz} & -T_{xxyz} & 0 & T_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & T_{yzyz} & T_{xxyz} \\ 0 & 0 & 0 & 0 & T_{xxyz} & \frac{T_{xxxx}}{2} - \frac{T_{xxyy}}{2} \end{bmatrix}$$

$$T_{xxxx} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} + 3\mathbb{T}_{40}^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)}$$
$$T_{xxyy} = \mathbb{T}_0^{(1)} + \mathbb{T}_{40}^{(1)} - 2\mathbb{T}_u^{(1)}$$
$$T_{xxzz} = \mathbb{T}_0^{(1)} - 4\mathbb{T}_{40}^{(1)} + \mathbb{T}_u^{(1)}$$
$$T_{xxyz} = \mathbb{T}_{4s}^{(1)}$$
$$T_{zzzz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} + 8\mathbb{T}_{40}^{(1)} + 4\mathbb{T}_u^{(1)} + 8\mathbb{T}_u^{(2)}$$
$$T_{yzyz} = \mathbb{T}_0^{(2)} - 4\mathbb{T}_{40}^{(1)} + \mathbb{T}_u^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & T_{xxzz} & T_{xxyz} & 0 & 0 \\ 0 & 0 & T_{xxzz} & -T_{xxyz} & 0 & 0 \\ -T_{xxzz} & -T_{xxzz} & 0 & 0 & 0 & 0 \\ -T_{xxyz} & T_{xxyz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -T_{xxyz} \\ 0 & 0 & 0 & 0 & T_{xxyz} & 0 \end{bmatrix}$$

$$T_{xxzz} = 3\mathbb{T}_u^{(3)}$$
$$T_{xxyz} = 2\mathbb{M}_{3c}^{(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} T_{yzyz} & 0 & 0 \\ 0 & T_{yzyz} & 0 \\ 0 & 0 & T_{xyxy} \end{bmatrix}$$

$$T_{yzyz} = \mathbb{T}_0^{(3)} - 2\mathbb{T}_u^{(4)}$$
$$T_{xyxy} = \mathbb{T}_0^{(3)} + 4\mathbb{T}_u^{(4)}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} T_{xxyz} & 0 & 0 \\ -T_{xxyz} & 0 & 0 \\ 0 & 0 & 0 \\ T_{yzyz} & 0 & 0 \\ 0 & -T_{yzyz} & 0 \\ 0 & -T_{xxyz} & 0 \end{bmatrix}$$

$$T_{xxyz} = \mathbb{M}_{3c}^{(2)}$$
$$T_{yzyz} = -3\mathbb{T}_u^{(5)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} T_{yzxx} & -T_{yzxx} & 0 & T_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & -T_{yzyz} & -T_{yzxx} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzxx} = \mathbb{M}_{3c}^{(3)}$$
$$T_{yzyz} = -3\mathbb{T}_u^{(6)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} T_{xxxx} & T_{xxyy} & T_{xxzz} & T_{xxyz} & 0 & 0 & T_{xxzy} & 0 & 0 \\ T_{xxyy} & T_{xxxx} & T_{xxzz} & -T_{xxyz} & 0 & 0 & -T_{xxzy} & 0 & 0 \\ T_{zzxx} & T_{zzxx} & T_{zzzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ T_{yzxx} & -T_{yzxx} & 0 & T_{yzyz} & 0 & 0 & T_{yzzy} & 0 & 0 \\ 0 & 0 & 0 & 0 & T_{yzzy} & T_{yzxx} & 0 & T_{yzyz} & T_{yzxx} \\ 0 & 0 & 0 & 0 & T_{xxzy} & \frac{T_{xxxx}}{2} - \frac{T_{xxyy}}{2} & 0 & T_{xxyz} & \frac{T_{xxxx}}{2} - \frac{T_{xxyy}}{2} \end{bmatrix}$$

$$
\begin{aligned}
T_{xxxx} &= \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} + 3\mathbb{T}_{40}^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} \\
T_{xxyy} &= \mathbb{T}_0^{(1)} + \mathbb{T}_{40}^{(1)} - 2\mathbb{T}_u^{(1)} \\
T_{xxzz} &= \mathbb{T}_0^{(1)} - 4\mathbb{T}_{40}^{(1)} + \mathbb{T}_u^{(1)} + 3\mathbb{T}_u^{(3)} \\
T_{xxyz} &= 2\mathbb{M}_{3c}^{(1)} + \mathbb{M}_{3c}^{(2)} + \mathbb{T}_{4s}^{(1)} \\
T_{xxzy} &= 2\mathbb{M}_{3c}^{(1)} - \mathbb{M}_{3c}^{(2)} + \mathbb{T}_{4s}^{(1)} \\
T_{zzxx} &= \mathbb{T}_0^{(1)} - 4\mathbb{T}_{40}^{(1)} + \mathbb{T}_u^{(1)} - 3\mathbb{T}_u^{(3)} \\
T_{zzzz} &= \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} + 8\mathbb{T}_{40}^{(1)} + 4\mathbb{T}_u^{(1)} + 8\mathbb{T}_u^{(2)} \\
T_{yzxx} &= -2\mathbb{M}_{3c}^{(1)} + \mathbb{T}_{4s}^{(1)} \\
T_{yzyz} &= \mathbb{T}_0^{(2)} - 4\mathbb{T}_{40}^{(1)} + \mathbb{T}_u^{(2)} - 3\mathbb{T}_u^{(5)} \\
T_{yzzy} &= \mathbb{T}_0^{(2)} - 4\mathbb{T}_{40}^{(1)} + \mathbb{T}_u^{(2)} + 3\mathbb{T}_u^{(5)}
\end{aligned}
$$

* Rank 4 tensor (a).

$$
\begin{bmatrix}
T_{yzxx} & -T_{yzxx} & 0 & T_{yzyz} & 0 & 0 & T_{yzzy} & 0 & 0 \\
0 & 0 & 0 & 0 & -T_{yzzy} & -T_{yzxx} & 0 & -T_{yzyz} & -T_{yzxx} \\
0 & 0 & 0 & 0 & 0 & T_{xyxy} & 0 & 0 & -T_{xyxy}
\end{bmatrix}
$$

$$
\begin{aligned}
T_{yzxx} &= \mathbb{M}_{3c}^{(3)} \\
T_{yzyz} &= \mathbb{T}_0^{(3)} - 2\mathbb{T}_u^{(4)} - 3\mathbb{T}_u^{(6)} \\
T_{yzzy} &= -\mathbb{T}_0^{(3)} + 2\mathbb{T}_u^{(4)} - 3\mathbb{T}_u^{(6)} \\
T_{xyxy} &= \mathbb{T}_0^{(3)} + 4\mathbb{T}_u^{(4)}
\end{aligned}
$$

* Rank 4 tensor (t).

$$
\begin{bmatrix}
T_{xxxx} & 0 & 0 \\
0 & T_{xxxx} & T_{yyyz} \\
0 & 0 & T_{zzzz} \\
0 & T_{yyyz} & T_{yyzz} \\
T_{zzxx} & 0 & 0 \\
0 & T_{xxyy} & -T_{yyyz} \\
0 & T_{yzzy} & 0 \\
0 & -T_{yyyz} & -\frac{T_{xxxx}}{2} + \frac{3T_{xxyy}}{2} + T_{yyzz} - T_{yzzy} + T_{zzxx} \\
\frac{T_{xxxx}}{2} - \frac{T_{xxyy}}{2} & 0 & 0 \\
-T_{yyyz} & 0 & 0
\end{bmatrix}
$$

$$
\begin{aligned}
T_{xxxx} &= 3\mathbb{T}_0^{(1)} + 3\mathbb{T}_{40}^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} \\
T_{yyyz} &= -\mathbb{T}_{4s}^{(1)} \\
T_{zzzz} &= 3\mathbb{T}_0^{(1)} + 8\mathbb{T}_{40}^{(1)} + 4\mathbb{T}_u^{(1)} + 8\mathbb{T}_u^{(2)} \\
T_{yyzz} &= \mathbb{T}_0^{(1)} - 4\mathbb{T}_{40}^{(1)} + \mathbb{T}_u^{(1)} + 3\mathbb{T}_u^{(3)} \\
T_{zzxx} &= \mathbb{T}_0^{(1)} - 4\mathbb{T}_{40}^{(1)} + \mathbb{T}_u^{(1)} - 3\mathbb{T}_u^{(3)} \\
T_{xxyy} &= \mathbb{T}_0^{(1)} + \mathbb{T}_{40}^{(1)} - 2\mathbb{T}_u^{(1)} \\
T_{yzzy} &= \mathbb{T}_0^{(1)} - 4\mathbb{T}_{40}^{(1)} + \mathbb{T}_u^{(2)} + 3\mathbb{T}_u^{(5)}
\end{aligned}
$$